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Fire-ball expansion and subsequent shock wave propagation from explosives detonation

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ABSTRACT

A well-known problem in thermo-hydrodynamics involves the sudden release of the explosion energy, concentrated in a finite volume, to surroundings with uniform density. However, only similarity solutions which have no detailed information on the earlier behavior of the fire-ball are available on this problem. In this study, we obtain a set of analytical solutions for the time dependent radius of an expanding fire-ball after explosives detonation by solving continuity, Euler and energy equations using a "polytrope" assumption for the fire-ball center. Subsequent spherical shock waves that develop from the fire-ball are obtained using the Kirkwood–Bethe hypothesis with Tait's equation of state for water and the ideal gas equation of state for air. The pressure waves emanating from the oscillating bubble in underwater, which has a notably different time scale from the shock wave generation, are obtained using the Rayleigh equation. The calculated results for the period and the maximum radius of bubble that resulted from the fire-ball and the pressure wave from the oscillating bubble in underwater are similar to the observed ones (Swift and Decius, 1946: Yennie and Arons, 1946). The calculated peak pressures as a function of shock radius in air medium were comparable to those observed (Swisdak, 1975).

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1. Introduction

With the detonation of a solid explosive, enormous amounts of energy are released and extremely high pressure develops while the solid density remains unchanged. After the detonation process terminates, the high pressure material expands rapidly and accompanied by a strong shock wave, known as ablast wave. Strong explosion accompanying shock waves in a homogeneous media has been thoroughly studied [1]. The distributions of pressure, density and gas velocity with respect to the shock radius were determined to solve the continuity and Euler equations with introducing a similarity variable by Sedov [1], which can also be found in Landau and Lifshitz [2]. The radius and velocity of the shock wave front were obtained by dimensional analysis [3]. However, the similarity solutions obtained in previous works do not adequately cover the region very near to the explosion source, i.e., they provide the velocity of the fire-ball only later time after the explosion.

Recently, numerical studies analyzing strong blast wave have been performed by several authors [4,5]. Especially, numerical simulation of underwater explosion was tried by various methods such as arbitrary Lagrangian–Eulerian formulation [6], indirect boundary element method [7], time-integration boundary-integral method [8] and adaptive solution technique [9]. In their study, the surrounding fluid was treated as either incompressible [4,7,8] or compressible [4,6,9] medium. However, the gas behavior inside the bubble was obtained by the relation of PV^n = constant. As is well known, the index n can range in the interval from 1 (isothermal) to the ratio of the specific heats γ (adiabatic) in polytropic approximation which assumes the uniform pressure and temperature for the gas inside the bubble intrinsically. Furthermore, the earlier stage of the expansion process which provides crucial information to obtain the subsequent shock propagation has never been treated properly by the numerical simulation [4,5]. In fact, initial radius and the corresponding pressure for the motion of the bubble evolved from the detonation products were chosen to match the observed bubble period [4]. On the other hand, with R_o and P_o which were taken as the initial conditions for the expansion of detonation products, the Rayleigh equation yields extremely large bubble period and the maximum radius, and consequently considerably large value in the shock strength.

The motion of the bubble evolved from the detonation products underwater is characterized by two different time scales having two orders of magnitude between the expansion of the detonation

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products and the pulsating of bubble [10]. It is really hard task to treat the underwater explosions involving several complex physical processes.

In this study, we obtained a set of analytical solutions for the time dependent radius of an expanding fire-ball after detonation of explosives by solving for the continuity, Euler and energy equations with a "polytrope" assumption at the fire-ball center. Spherical shock waves that developed from the rapidly expanding fire-ball were obtained using the Kirkwood–Bethe hypothesis [11] with Tait's equation of state for water and with the ideal gas equation of state for air medium. Subsequent pressure waves emanating from the oscillating bubble in underwater are obtained using the Rayleigh equation [12]. In an air environment, the Rayleigh equation, neglecting the pressure term, was used to calculate the behavior of the fire-ball after rapid expansion.

2. Continuity and the Euler equation and their solutions

The equation for continuity and the Euler equation for an irrotational fluid are given as

$$\frac{\partial \rho}{\partial t} + \nabla \cdot \left(\rho \vec{u}\right) = 0 \tag{1}$$

$$\frac{\partial \vec{u}}{\partial t} + \vec{u} \cdot \nabla \vec{u} = -\frac{1}{\rho}\nabla P \tag{2}$$

Here $\vec{u}$ is the fluid velocity. With decomposition of the gas density in spherical symmetry as $\rho = \rho_o(t) + \rho_r(r,t)$ the continuity equation becomes

$$\left[\frac{\partial \rho_o}{\partial t} + \rho_o \nabla \cdot \vec{u}\right] + \frac{D\rho_r}{Dt} + \rho_r \nabla \cdot \vec{u} = 0 \tag{3}$$

where the notation of the total derivative used here is $D/D = \partial/\partial t + u\partial/\partial r$.

The rate of change of the density of a material particle can be represented by the rate of volume expansion of that particle in the limit [13] $V \to 0$, or

$$\nabla \cdot \vec{u} = \lim_{V \to o} \left(\dot{V}/V\right) = 3\dot{R}/R \tag{4}$$

With this result, one can have radial dependent velocity profile inside the fire-ball from the continuity equation.

$$u = \dot{R}r/R \tag{5}$$

This linear velocity means that the fire-ball expands homologously [14]. In other words, every mass point during expansion may be traced back to a single point, the center of the fire-ball. With this velocity profile, one can obtain the following quadratic density profile. The fully detailed derivation can be found in Kwak and Yang [15] and Kwak and Jun [16].

$$\rho(r,t) = \frac{b}{R^3(t)} + \frac{ar^2}{R^5(t)} \tag{6}$$

where"a" and "b" are constants related to the mass of the fire-ball.

Substituting Eqs. (5) and (6) into the Euler equation given in Eq. (2), we have the following pressure profile inside the fire-ball:

$$P = P_0(t) - \left(\frac{b}{2}\frac{r^2}{R^2} + \frac{a}{4}\frac{r^4}{R^4}\right)\frac{\ddot{R}}{R^2} \tag{7}$$

At the center of the fire-ball, the above equation with a boundary condition, P = 0 at r = R becomes:

$$P_0 = \left(\frac{a+2b}{4}\right)\frac{\ddot{R}}{R^2} \tag{8}$$

To determine the "a" and "b" in Eq. (6), we need an equation of state for the gas inside the fire-ball. Excluding the viscous dissipation term, the energy equation with internal and enthalpy representation can be written as [15,17]

$$\rho C_v \frac{DT}{Dt} = -P\nabla \cdot \vec{u} - \nabla \cdot \vec{q} \tag{9}$$

$$\rho C_p \frac{DT}{Dt} = \frac{DP}{Dt} - \nabla \cdot \vec{q} \tag{10}$$

where C_v andC_p are the specific heats at constant-volume and at constant-pressure, respectively, and $\vec{q}$ is the heat flux. Eliminating$\nabla \cdot \vec{q}$ from Eqs. (9) and (10) give us

$$\rho\left(C_p - C_v\right)\frac{DT}{Dt} = \rho R_g \frac{DT}{Dt} = P\nabla \cdot \vec{u} + \frac{DP}{Dt} \tag{11}$$

where R_g is the gas constant. At the center of the fire-ball, the above equation reduces to

$$\rho_0 R_g \frac{\partial T_0}{\partial t} = 3P_0\frac{\dot{R}}{R} + \frac{\partial P_0}{\partial t} \tag{12}$$

which yields the following ideal gas relation at the center of the fire-ball.

$$P_o R^3/T_o = const. \tag{13}$$

whereP_o and T_o are the gas pressure and temperature, respectively, at the fire-ball center.

The adiabacity condition at the fire-ball center can be written as:

$$P_0 = \kappa \rho_0^{\gamma_T} \tag{14}$$

whereγ_T is the specific heat ratio for the detonation products of the explosives and κ is some constant.

The assumption of the polytrope at the center is equivalent to the assumption that the center is neither a heat source nor a heat sink, i.e., $\nabla \cdot \vec{q} = 0$ at the center. When the velocity, density and pressure profiles are obtained, and are given in Eqs. (5–7), we can solve the energy equation with the boundary condition given in Eq. (14) to obtain the following results.

$$a = -b = -15M/8\pi \tag{15}$$

$$\rho(r,t) = \frac{b}{R^3(t)}\left[1 - \left(\frac{r}{R}\right)^2\right] \tag{16}$$

$$T(r,t) = T_s + \frac{R\ddot{R}}{4R_g}\left[1 - \left(\frac{r}{R}\right)^2\right] \tag{17}$$

$$P(r,t) = \frac{b}{4}\frac{\ddot{R}}{R^2}\left[1 - \left(\frac{r}{R}\right)^2\right]^2 \tag{18}$$

where M is the mass of the explosive.

Note that even though the temperature at the surface of the fire-ball T_s is included in Eq. (17), it cannot be calculated using the ideal gas law because the gas pressure at the surface of the fire-ball becomes null from Eq. (18), which indicates that the properties at the surface of the fire-ball cannot be defined. In fact, the pressure, density and gas velocity distributions behind the shock front may be obtained from Sedov analysis [1].

With the assumption of polytrope at the center of the fire-ball, which is given in Eq. (14) and the solutions given in Eqs. (15) and (16), the equation for the pressure at the center of the fire-ball becomes the following equation for the motion of the fire-ball.

$$\frac{d^2}{dt^2}R(t) = \frac{32\pi\kappa'}{15M}\frac{1}{R^{3\gamma_T-2}} \tag{19}$$

where $\kappa' = P_0R^{3\gamma_T}$.

Thus the continuity equation, the Euler equation and the energy equation with the assumption of polytrope only at the center can be reduced to Eq. (19), which can be solved using the standard "energy" method. After multiplying by $\dot{R}$ to Eq. (19) and integrating, we have typical equation for the energy conservation.

$$\frac{1}{2}\dot{R}^2 + W(R) = \varepsilon \tag{20}$$

where ε is total energy and the potential $W(R)$ is given by

$$W(R) = \frac{32\pi\kappa'}{45M(\gamma_T-1)}\frac{1}{R^{3\gamma_T-3}} \tag{21}$$

The solution of Eq. (19) and hence Eq. (20) is then given by an elementary integral such as

$$\int_{R_0}^{R}\frac{dR}{\sqrt{2[\varepsilon - W(R)]}} = t \tag{22}$$

If the specific heat ratio γ_T has a specific value, the integral in Eq. (22) can be written explicitly, i.e. for$\gamma_T = 5/3$, and the solution for Eq. (22) can be expressed as

$$R = R_0\cosh\theta \tag{23}$$

$$t = \sqrt{\frac{45M(\gamma_T-1)R_0^4}{64\pi\kappa'}}\sinh\theta \tag{24}$$

where R_o is the initial radius of the fire-ball.

For $\gamma_T = 4/3$, the solution for the equation of motion is given by

$$R = R_0\cosh^2\theta \tag{25}$$

$$t = \sqrt{\frac{45M(\gamma_T-1)R_0^3}{64\pi\kappa'}}\left[\frac{\sinh(2\theta)}{2}+1\right] \tag{26}$$

From Eqs. (23) and (24), and also from Eqs. (25) and (26), one can obtain the asymptotic value of the fire-ball velocity as:

$$\frac{dR}{dt} = 0.89\sqrt{2E'/M} \tag{27}$$

where $E' = P_oV_o$ andV_o is the initial volume of TNT.

On the other hand, Sedov obtained the following pressure profile and asymptotic velocity of the fire-ball with a constant density assumption [1]:

$$P(r,t) = P_o(t)\left[1-\left(\frac{r}{R}\right)^2\right] \tag{28}$$

$$\frac{dR}{dt} = \left(\frac{5}{3}\right)^{1/2}\sqrt{2E_{\exp}/M} \tag{29}$$

where $E_{\exp}$ is the internal energy of the explosive.

3. Normalized solutions

We normalized the solutions using the following non-dimensional parameters [18]:

$$R^* = R/(E_{\exp}/P_\infty)^{1/3} \tag{30}$$

$$t^* = tc_\infty/(E_{\exp}/P_\infty)^{1/3} \tag{31}$$

$$P^* = P/P_\infty \tag{32}$$

where c_∞ is the sound velocity in the medium and P_∞ is the ambient pressure.

Even though the value of the specific heat ratio $\gamma_T = 4/3$ is more appropriate for TNT products, the value of $\gamma_T = 5/3$ was chosen in this study because it is easy to get $\theta - t$ relation from Eq. (24). With these non-dimensional parameters, the solution for the fire-ball with $\gamma_T = 5/3$ can be given as:

$$R^* = R_0^*\cosh\theta \tag{33}$$

$$t^* = \sqrt{\frac{30}{64\pi P_0^*R_0^*E_{\exp}^*}}\sinh\theta \tag{34}$$

where the initial energy of the explosive$E_{\exp}$ is normalized by Mc_∞^2.

$$E_{\exp}^* = E_{\exp}/\left(Mc_\infty^2\right) \tag{35}$$

Then the instantaneous velocity of the fire-ball can be obtained from Eqs. (33) and (34).

$$\frac{dR^*}{dt^*} = R_0^*\sqrt{\frac{64\pi P_0^*R_0^*E_{\exp}^*}{30}}\tanh\theta \tag{36}$$

Non-dimensional acceleration of the fire-ball can be obtained by differentiating the above equation w.r.t. t^*. The instantaneous velocity depends only on initial TNT properties such as gas energy, gas pressure and the radius of the fire-ball. After a fraction of ms, Eq. (36) having a constant velocity at asymptotic limit is no longer valid since the acceleration of the fire-ball wall and the gas pressure inside the fire-ball reduce to the extent that the fire-ball velocity suddenly drops to nearly zero, which can be confirmed from the numerical simulation [6].

As the velocity of the fire-ball decreases nearly zero, the nonlinear behavior of the fire-ball begins [19]. The subsequent expansion of the fire-ball in underwater can be described using the following Rayleigh equation. However, it is not easy task to match the fire-ball expansion by Eq. (34) whose characteristic time is tens of micro-second to the Rayleigh equation having characteristic time of millisecond in this case. In this study, a criterion for this transition was taken as the non-dimensional acceleration of the fire-ball.

$$R\ddot{R} + 3\dot{R}^2/2 = (P - P_\infty)/\rho_\infty \tag{37}$$

During the nonlinear motion of the fire-ball, the temperature and pressure inside the fire-ball are assumed to be uniform. In an air medium, the pressure term in Eq. (37) may be negligible so that the equation for the fire-ball after the initial expansion may be written as [20]

$$R^3\dot{R}^2 = const. \tag{38}$$

The relationship above yields $R \propto t2/5$, which can be obtained from Taylor's non-dimensional analysis [3,20].

4. Shock wave propagation

A rapidly expanding fire-ball may emit shock waves outward. The velocity and the pressure fields in liquid, caused by the detached shock waves can be obtained using the Kirkwood–Bethe hypothesis. This hypothesis assumes that the invariant quantity Y propagates in the medium with the characteristics velocity $c + u$ [11]:

$$\left[\frac{\partial}{\partial t} + (c+u)\frac{\partial}{\partial r}\right]Y = 0 \tag{39}$$

The invariant Y is defined as follows.

$$Y = r\left(h + \frac{u^2}{2}\right) = R\left(H + \frac{U^2}{2}\right) \tag{40}$$

A relationship between enthalpy to the local sound velocity and pressure is obtained by the following relationship for the isentropic relation.

$$h = \int_{p_\infty}^{p} \frac{dp}{\rho} \tag{41}$$

Thus, the continuity equation for a spherical coordinate can be written as

$$-\frac{1}{c^2}\frac{Dh}{Dt} = \frac{\partial u}{\partial r} + \frac{2u}{r} \tag{42}$$

The outgoing shock characteristics can be obtained using Eq. (39) and the continuity equation for fluid [11]. These are given for water, as [21]

$$\frac{dr}{dt} = c + u \tag{43}$$

$$\left.\frac{du}{dt}\right)_{char} = \frac{1}{c-u}\left[(c+u)\frac{Y}{r^2} - \frac{2c^2u}{r}\right] \tag{44}$$

$$\left.\frac{dp}{dt}\right)_{char} = \frac{\rho_\infty}{r(c-u)}\left(\frac{p+B'}{p_\infty + B'}\right)^{1/n}\left[2c^2u^2 - \frac{c(c+u)}{r}Y\right] \tag{45}$$

The values of Y, which can be obtained from the instantaneous fire-ball motion are necessary to solve the aforementioned shock characteristics.

Accounting to the above equations, we assumed that the density of the liquid is a function of pressure only and that the compression process of the liquid is isentropic. The appropriate equation of state for a liquid suited to these circumstances is Tait equation, which is given by [11]:

$$\frac{p+B'}{p_\infty + B'} = \left(\frac{\rho}{\rho_{\infty,w}}\right)^n \tag{46}$$

For water, B' is approximately 304.0 MPa and n is equal to 7. The local sound velocity, c_w in water with instantaneous liquid pressure, p can be obtained using the above equation of state.

$$c_w = c_{\infty,w}\left(\frac{p+B'}{p_\infty + B'}\right)^{\frac{n-1}{2n}} \tag{47}$$

One can obtain the enthalpy at the fire-ball H_w at the wall by integrating Eq. (41) using the Tait equation. H_w has the following form.

$$H_w = \frac{n}{n-1}\frac{(p_\infty + B')}{\rho_{\infty,w}}\left[\left(\frac{p+B'}{p_\infty + B'}\right)^{\frac{n-1}{n}} - 1\right] \tag{48}$$

In an air medium, the gas compression during the development of the shock wave was assumed to be adiabatic, that is, the relation holds $p\rho^{\gamma_a} = const.$ so that the local sound velocity c_a and the enthalpy at the fire-ball wall, which correspond to Eqs. (47) and (48), respectively, are given as follows [21,22].

$$c_a = c_{\infty,a}\left(\frac{p}{p_\infty}\right)^{\frac{\gamma_a - 1}{2\gamma_a}} \tag{49}$$

$$H_a = \frac{c_{\infty,a}^2}{\gamma - 1}\frac{p_\infty}{\rho_{\infty,a}}\left[\left(\frac{p}{p_\infty}\right)^{\frac{\gamma_a - 1}{\gamma_a}} - 1\right] \tag{50}$$

where γ_a the specific heats ratio for air.

The shock characteristics given in Eqs. (43) and (44) can be used in an air medium. However, the shock characteristics in water, Eq. (45) should be replaced as the following equation for air medium.

$$\left.\frac{dp}{dt}\right)_{char} = \frac{\rho_{\infty,a}}{r(c-u)}\left(\frac{p}{p_\infty}\right)^{1/\gamma_a}\left[2c^2u^2 - \frac{c(c+u)}{r}Y\right] \tag{51}$$

Eqs. (43–45) for the shock characteristics underwater, and Eq. (43), (44) and (51) for those of in air medium, provide the thickness D of the shocked gas layer. The thickness of the shock layer may be obtained with assumption that all the mass inside the fire-ball is swept into the layer [1,20]. That is given by

$$D = \frac{1}{3}\left(\frac{\gamma - 1}{\gamma + 1}\right)r_s \tag{53}$$

where r_s is the shock radius.

5. Results and discussion

The gaseous properties of the detonation products of TNT were obtained using the Jones–Wilkins–Lee (JWL) equation of state [6], resulting in a density of 1630 kg/m^3, a specific internal energy of 4.29×10^6 J/kg and a pressure of 8.15×10^9 Pa for TNT. The temperature of the detonation products was taken as 3300 K. In fact, the temperature estimated by the Chapman–Jouguet state with the JWL equation of state for the TNT explosive was approximately 3643 K. The initial radius of the detonation products was calculated with a relationship of $\rho V_o = M$, where V_o was the initial volume of the detonation products.

A schematic for the formation of shell layer with the shocked gas due to the rapid expansion of the detonation products [1,20] is

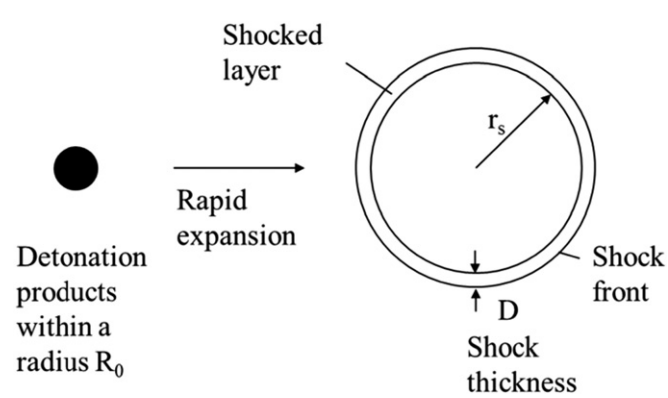

Fig. 1. Schematic representation of shock generation due to rapid expansion of detonation products after complete explosion (Ref. [20]).

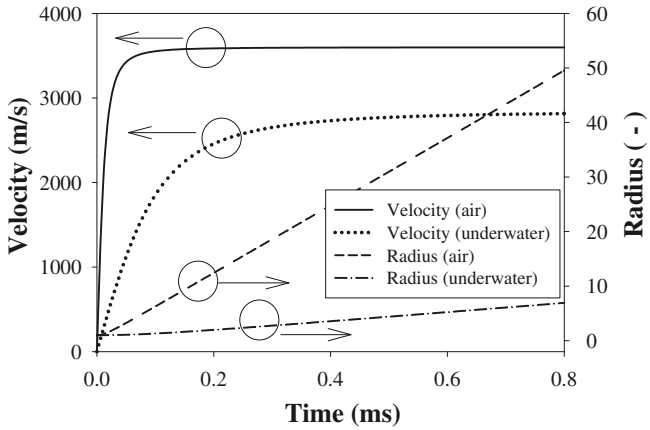

Fig. 2. Time dependent velocity and radius of the fire-ball in air and underwater.

displayed in Fig. 1. Fig. 2 shows the calculated values of time dependent velocity and non-dimensional radius of the fire-ball for 250 kg of TNT which exploded at a depth of 10 m underwater. Initial radius of the detonation products was taken as 0.3319 m in this case. After the detonation occurred, the fire-ball velocity increased linearly and reached its asymptotic limit of 2810 m/s after less than a fraction of ms. However, the velocity of the fire-ball increases more rapidly to reach an asymptotic value of 3560 m/s after within 50 μs in air medium. The asymptotic velocity of the fire-ball in air medium is slightly less than the value of 3780 m/s obtained by Eq. (29). As shown in Fig. 3, the center pressure and temperature which are obtained from Eqs. (13) and (18), respectively, reduced very abruptly in underwater as the radius of the fire-ball increased linearly. More rapid decrease of pressure and temperature at the center of the fire-ball can be expected in air because the shock radius increased more rapidly in air.

Propagation of shock front due to the rapid expansion of the products by the detonation of a 250 kg of TNT in underwater is shown in Fig. 4. Integration of Eqs. (43–45) using the 4th order Runge–Kutta numerical method yielded the instantaneous the shock distance propagated, particle velocity and shock strength, respectively. In these integrations, the instantaneous value of Y defined in Eq. (40) was used as an initial condition to obtain the shock characteristics. The peak pressure of the shock front is about 7305.5 MPa at 6 μs and 6505.1 MPa at 12 μs after the detonation. The thickness of the shell layer of the shock front during this period is about 0.5 mm which is less than the value estimated by Eq. (53)

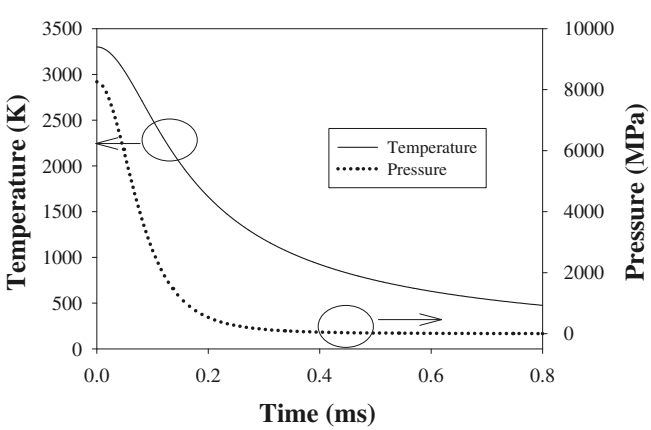

Fig. 3. Time dependent temperatures and pressure at the center of the fire-ball underwater.

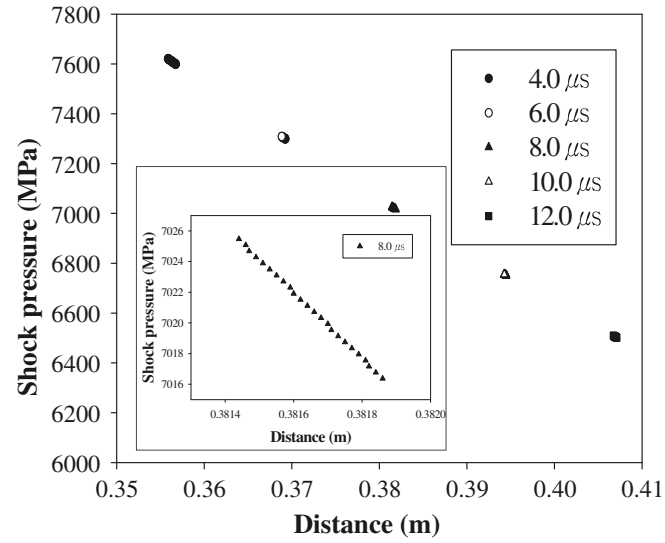

Fig. 4. Propagation of shock front due to the rapid expansion of detonation products from a 250 kg TNT charge at a depth of 10 m underwater.

by an order of magnitude. The over-estimation for the shocked layer by Eq. (53) indicates that all the gas inside the fire-ball cannot be swept into the layer. The pressure difference between the outer and inner layer of the shock front is about 9.1 MPa. The maximum propagation velocity of the shock front is approximately 6250 m/s during this period. A group of points in Fig. 4 indicates the shell layer of the shock front and a detailed example is shown in the insert. Fig. 5 shows the propagation of the shock front around a 250 kg of TNT spherical charge at several specific times after a complete detonation. The peak pressure of the spherical shock fell off more rapidly with increased distance than did 1/r [23] which is true for a propagation of a small amplitude shock. The peak magnitude of the shock front at 1.6 m from the fire-ball center is about 759.9 MPa. The thickness of the shell layer of the shock front during this stage is about 10 mm. Shock front propagates with velocity of 6350 m/s in this period.

When the acceleration of the wall of the fire-ball and the gas pressure at the center of the fire-ball was reduced, the velocity of

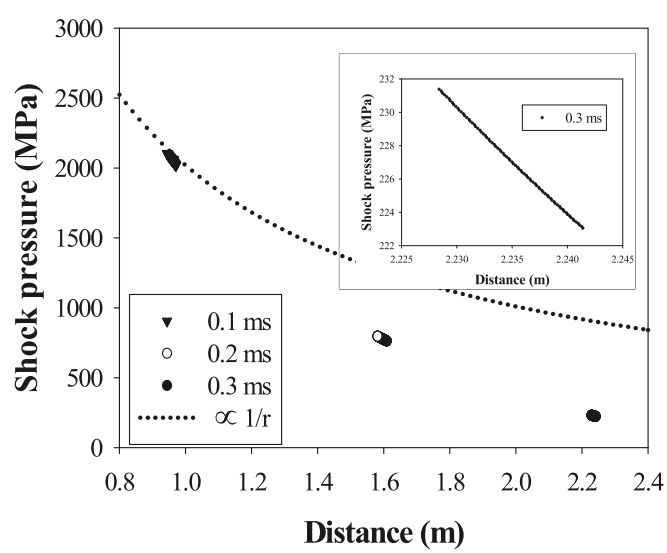

Fig. 5. Propagation of shock front at several specific times after detonation completion from a 250 kg TNT charge underwater.

Table 1
Period and maximum radius of a fire-ball evolved from a 0.3 kg TNT charge at a depth of 91.4 m underwater dependent on the initial conditions of the non-dimensional acceleration value.

Non-dimensional acceleration (fire-ball radius , m/pressure, MPa)	Period of bubble developed from the fire-ball (ms)		Maximum radius (m)	
	Measured value	Calculated value	Measured value	Calculated value
0.09 (0.327/1.11)	29.8	29.0	0.481	0.437
0.07 (0.354/0.80)		28.2		0.452
0.05 (0.393/0.53)		27.8		0.482
0.04 (0.423/0.40)		28.0		0.510
0.03 (0.464/0.23)		28.8		0.552

the fire-ball becomes nearly zero where the nonlinear behavior of the fire-ball started [19]. However, it was very difficult to determine the initial conditions for the nonlinear behavior of the fire-ball, because the nonlinear motion of the fire-ball (or bubble) after the shock emission crucially depends on the initial conditions that are chosen. In this study, a specific value for the non-dimensional acceleration of the fire-ball was chosen as an initial condition to determine the subsequent nonlinear motion of the fire-ball. The pulsating behavior of the bubble was calculated by the Rayleigh equation using the 4th order Runge–Kutta numerical method. In Table 1, the periods and the maximum size of the fire-ball evolved from a detonation of 0.3 kg of a TNT spherical charge at a depth of 91.4 m with various initial conditions for the acceleration and corresponding radius and pressure of the fire-ball are given. As can be seen in Table 1, a non-dimensional acceleration value of 0.05 provided a reasonable value for the maximum radius compared to the measured values for the 0.3 kg TNT charge [24]. If one chooses the initial point for the nonlinear motion as the non-dimensional acceleration of 0.05,the velocity of the fire-ball at the starting point of the nonlinear motion turned out to be about 19.2 m/s for all of the cases listed in Table 1.

Fig. 6 shows a radius–time curve of the fire-ball for the case of 0.3 kg TNT charge exploded at 91.4 m underwater. The collapsing phase of the fire-ball is not a replica of the expanding phase of the solid explosive, because the characteristic time for the expanding process of the fire-ball is microseconds, whereas the characteristic time for the collapsing phase governed by the Rayleigh equation is mili-seconds. Notable deviation between our result and the calculation result by Luo et al. [6] occurs at the initial phase of expanding process. A proper match between the initial expansion process and the pulsating motion of the bubble will provides more reasonable result, which needs further study.

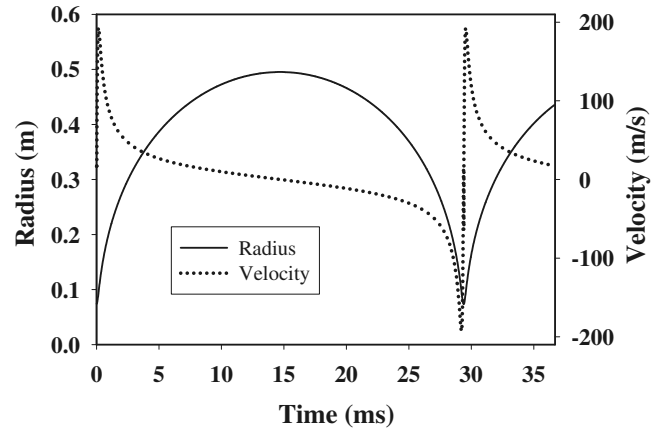

Fig. 7. Time dependent radius and velocity of a 0.3 kg TNT charge calculated by the Rayleigh equation.

With proper initial radius and the corresponding gas pressure, the Rayleigh equation provides the bubble radius–time curve calculated by Luo et al. The time radius curve and velocity calculated by the Rayleigh equation for the fire-ball of 0.3 kg charge is displayed in Fig. 7. Initial radius and pressure used in this calculation are 0.0741 m and 199.6 MPa, respectively. The initial radius chosen is 2.1 times of the charge radius and the initial pressure was estimated by $PR^{3\gamma}$ relation from the radius and pressure of 0.3 kg charge. Quite close agreement between Luo's results and our calculations can be seen. If one chooses the initial conditions as charge radius and pressure which are 0.0353 and 8150 MPa, respectively, 60% increase in the maximum radius and period of oscillation is obtained. In fact, an initial radius of the fire-ball having 1.8 times the charge radius was found to produce reasonable period of the fire-ball [4]. However one may lose information on the shock propagation in this calculation.

We also computed the far-field pressures from the fire-ball and compared them with the experimental results by Yannie and Arons [25] (Fig. 8). This test was performed using a 0.227 kg of TNT at

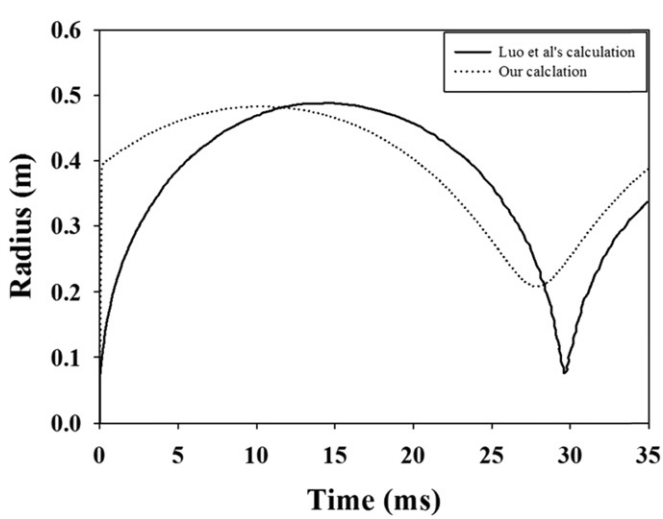

Fig. 6. Fire-ball radius time for a 0.3 kg TNT spherical charge at a depth of 91.4 m underwater

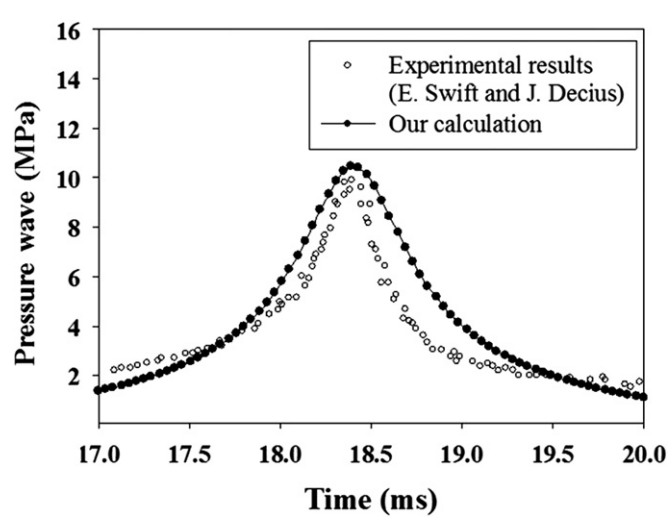

Fig. 8. Calculated and observed results for the far-field pressures at 0.695 m from a 227 g TNT charge at a depth of 152.4 m underwater.

Table 2
Calculated and measured values.

Depth (m)/Mass of TNT (kg)	Period of bubble developed from the fire-ball (ms)		Maximum radius (m)	
	Measured value	Calculated value	Measured value	Calculated value
91.4/0.30	29.8	27.9	0.481	0.483
152.4/0.227	18.3	16.9	0.379	0.406
178.6/0.30	17.8	16.4	0.387	0.439
91.7/28.0	(127.0)*	126.0	(2.09)*	2.19
10.0/250.0	–	1063.2	–	7.30

*The values in parenthesis are from the numerical method (Ref. [6]) and the measured values are from Refs. [24,25].

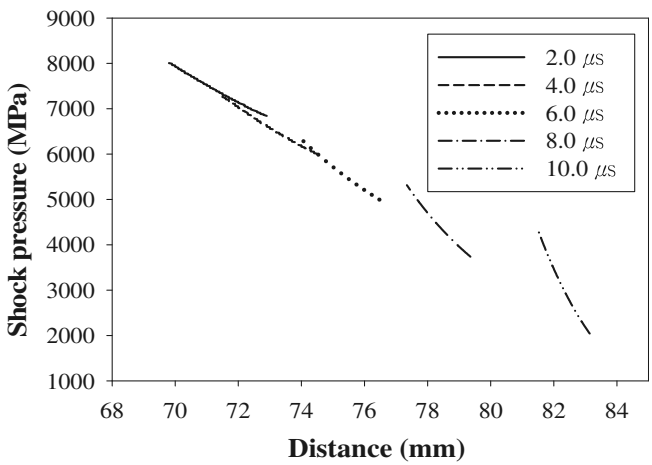

Fig. 9. Shock generation due to the rapid expansion of detonation products of 1 kg of TNT charge in air.

a depth of 152.5 m and the pressure was monitored at a location 0.695 m from the center of the charge. The initial radius of 0.0321 m was taken in this case. The calculated values were close to the observed values. Table 2 details the comparison between the calculation results and the test results by Swift and Decius (0.3 kg of TNT) [24] and Yennie and Arons [25] for the bubble period. The maximum radius is also shown there. The observed and calculated results were in close agreement.

Fig. 9 shows the propagation of the shock front due to the linear expansion of detonation products, i.e. the linearly increase in the fire-ball radius as shown in Fig. 2, in an air medium. The peak pressure of the shock front decreased very rapidly as the shock front moved. The thickness of the shell layer of the shock front is about 3 mm which is comparable to the estimated value by Eq. (53). The pressure difference between the outer and inner layer of the shock front is as large as 1165.2–2259.5 MPa during this stage. The spherical shocks at several specific times shown in Fig. 10 are due to the nonlinear expansion of the fire-ball, which is described in Eq. (38). The figure in the insert shows the typical pressure profile for the spherical shock in air. The shock speed during this period is approximately 4200 m/s. The calculated peak pressures as a function of the shock radius are similar to the observed ones [26].

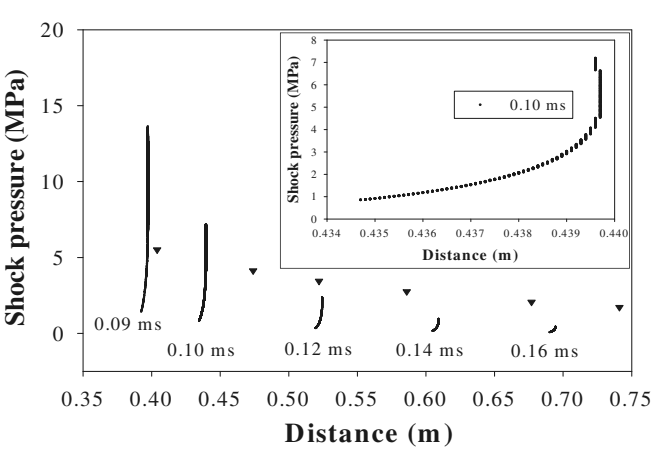

Fig. 10. Shock wave propagation at several specific times after completion of detonation from a 1 kg of TNT charge in air. Dark triangles indicate the experimental values by Swisdak.

6. Conclusions

A set of analytical solutions for an expanding fire ball developed by the detonation of an explosive was obtained by solving continuity, Euler and energy equations. The initial phase of the explosion which was first obtained in this study turned out to be very important to understand subsequent shock propagation into medium. The spherical shock driven by the explosive was obtained using the Kirkwood–Bethe hypothesis. Shock generation due to rapidly expanding detonation products underwater and in air medium was calculated numerically. The calculated values for the period and maximum size of the bubble developed from the fire-ball were similar to the observed values underwater. Moreover, the calculated shock pressures as a function of the shock radius also agree with the observed values in air. An approach for the shock propagation due to the detonation of explosives in this study may be used as complement to the Sedov's work [1] which provides the distribution of pressure, density and velocity behind the shock front.

Acknowledgements

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Nomenclature

c	sound velocity
C_p	coefficient of specific heat at constant pressure
C_v	coefficient of specific heat at constant volume
D	shock thickness
$E_{\exp}$	internal energy of explosive
h	enthalpy
H	enthalpy at the fire-ball wall
M	mass of the fire-ball and mass of explosive
p	shock pressure
P	gas pressure inside the fire-ball
q	heat flux
r	distance from fire-ball center
r_s	shock radius
R	radius of fire-ball
$\dot{R}$	velocity of the fire-ball
$\ddot{R}$	acceleration of the fire-ball

R_o	initial radius of the fire-ball
R_g	gas constant
t	time
T	gas temperature inside fire-ball
u	velocity inside fire-ball
U	velocity at the fire-ball wall
u_s	shock velocity
V	volume of the detonation products
W	potential energy of fire-ball

Greeks

ε	total energy of fire-ball
γ	specific heat ratio of gas inside fire-ball
κ	a constant defined in Eq. (14)
ρ	gas density inside fire-ball
ρ_∞	density in medium
θ	a parameter

Superscripts

*	non-dimensional parameter
$\cdot$	derivative with respect to time
$\cdot\cdot$	second derivative with respect to time

Subscripts

a	air
$char$	characteristics of shock wave
o	center
r	radial dependence
s	shock or surface
T	TNT
w	water and the fire-ball wall
∞	medium

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